

INTRODUCTION TO STATISTICAL METHODS

Gaussian Mixture Models

Soft clustering, the mixture model, responsibilities, and the EM algorithm.

Companion reader · Read beside the slides

How to use this companion. Colour boxes guide you: **blue** intuition, **amber** everyday picture, **green** worked example, **red** watch out, **purple** key takeaway. Read it next to the slides, in order.

1 From hard clusters to soft ones

k-means gives every point a *hard* label — it belongs fully to one cluster. But real points near a boundary are genuinely ambiguous. A **Gaussian Mixture Model** (GMM) does **soft clustering**: it models the data as coming from a mixture of several **Gaussian** (normal) distributions, and assigns each point a *probability* of belonging to each cluster.

The intuition

Instead of drawing hard fences, a GMM lays down a few overlapping “bell-shaped hills”. A point sitting on the slope between two hills is, say, 80% from one and 20% from the other — a much more honest description than forcing it into one.

2 The mixture model

A GMM with K components says the density of the data is a weighted sum of Gaussians:

$$p(\mathbf{x}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x} \mid \mu_k, \Sigma_k),$$

where π_k are the **mixing weights** ($\sum_k \pi_k = 1$), and each component has its own mean μ_k and variance/covariance Σ_k . Fitting means finding the π_k, μ_k, Σ_k that best explain the data.

3 Responsibilities: the soft assignment

For a point $\mathbf{x}$, its **responsibility** to cluster k is the posterior probability that it came from that component:

$$r_k(\mathbf{x}) = \frac{\pi_k \mathcal{N}(\mathbf{x} \mid \mu_k, \Sigma_k)}{\sum_j \pi_j \mathcal{N}(\mathbf{x} \mid \mu_j, \Sigma_j)}.$$

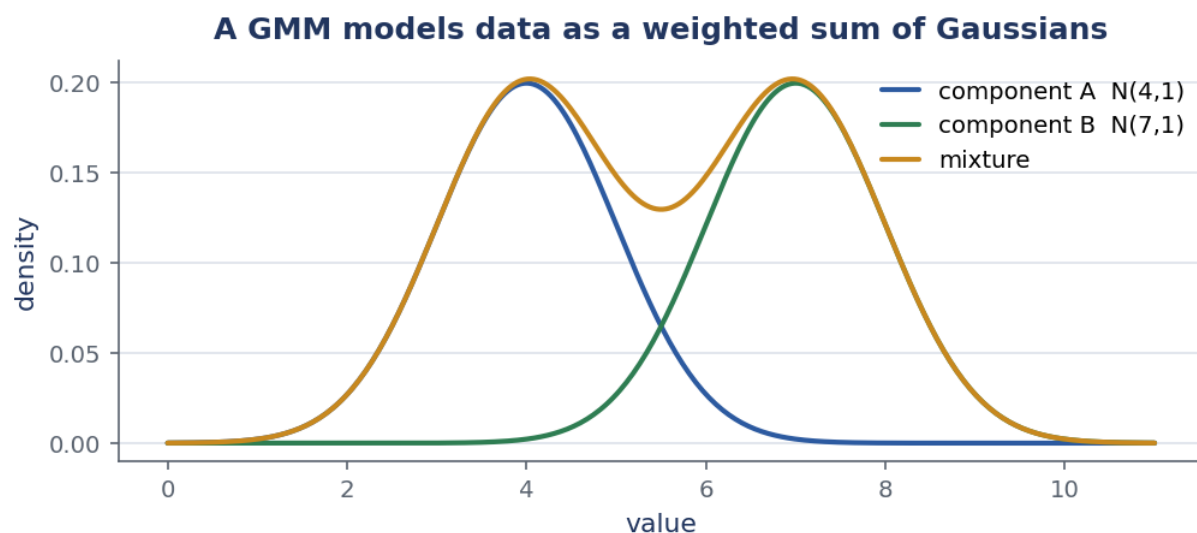


Figure 1: A GMM represents the data as a weighted sum of Gaussians; the mixture (top curve) is what you observe, and the components are the hidden clusters.

Worked example: computing a responsibility

Two equally-weighted 1-D clusters, $A = \mathcal{N}(4, 1)$ and $B = \mathcal{N}(7, 1)$, $\pi_A = \pi_B = 0.5$. For the point $x = 5$: $\mathcal{N}(5 | 4, 1) = \frac{e^{-0.5(1)^2}}{\sqrt{2\pi}} = \mathbf{0.242}$ and $\mathcal{N}(5 | 7, 1) = \frac{e^{-0.5(2)^2}}{\sqrt{2\pi}} = \mathbf{0.054}$. So

$$r_A = \frac{0.5(0.242)}{0.5(0.242) + 0.5(0.054)} = \frac{0.242}{0.296} = \mathbf{0.818}.$$

The point is $\approx 82\%$ cluster A , 18% cluster B — a soft, probabilistic assignment (k-means would just say “A”).

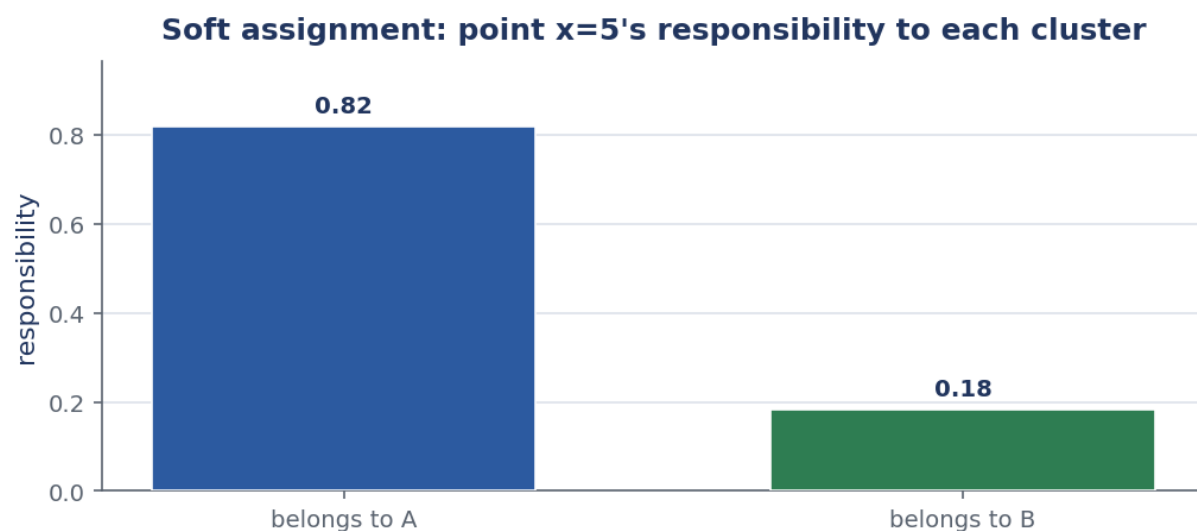


Figure 2: Soft assignment: $x = 5$ is 82% responsible to A and 18% to B rather than being forced entirely into one cluster.

4 Fitting with EM

GMMs are fit by the **Expectation–Maximisation** (EM) algorithm, which alternates two steps like a soft version of k-means:

- **E-step:** with current parameters, compute each point’s responsibilities r_k .
- **M-step:** update each cluster’s weight, mean and variance as the responsibility-weighted averages of the data.

Repeat until the parameters stop changing. (k-means is the special case where clusters are spherical, equal size, and assignments are forced to 0/1.)

✓ Key takeaway

A **GMM** models data as a weighted sum of Gaussians $\sum_k \pi_k \mathcal{N}(\mu_k, \Sigma_k)$ and does *soft* clustering: each point gets a **responsibility** r_k per cluster (worked: $x=5$ is 82% A, 18% B). It is fit by **EM** — E-step computes responsibilities, M-step updates weighted means/variances — generalising k-means (hard, spherical) to overlapping, differently-shaped clusters.